

HealthConnect – Week 6 Project Summary

Integration, Advanced Development & Validation

Project Overview

HealthConnect is focused on reducing missed appointments, improving attendance, making better use of appointment slots, and strengthening administrative support through data and AI. Week 6 continues the Week 5 work by improving, integrating and validating existing outputs rather than starting a new project.

Week 6 Focus

The Week 6 progression is: Week 5 Output → Improve → Integrate → Validate → Document → Prepare for Week 7 testing. The Project Management track is responsible for project integration and coordination across workstreams.

Track Progress and Key Outputs

- **Data Analytics:** Conducted deeper analysis and validated key performance indicators, focusing on booking lead time, previous no-show history, reminder status and channel, appointment type, distance, and patient segments. The team is also refining the dashboard and developing evidence-based operational recommendations.
- **Data Science:** Conducted formal error analysis of false positives and false negatives, addressed feature redundancy, and tuned a Gradient Boosting model. Model performance improved from the Week 5 baseline of 62.4% accuracy and 0.677 ROC-AUC to 64.8% accuracy and 0.687 ROC-AUC.
- **ML Engineering:** Integrated preprocessing, feature engineering, and the candidate model into a reusable pipeline. This included schema validation, batch prediction, configurable risk thresholds, model save/reload functionality, and end-to-end integration testing, with 10/10 tests successfully passed.
- **Generative AI:** Focused on evaluating and refining the HealthConnect assistant as part of Week 6 development. Progress, issues, dependencies, and integration outcomes will be documented as the GenAI work is finalized.

Cross-Track Integration

Meaningful integration is being coordinated between Data Analytics and Data Science, Data Science and ML Engineering, Analytics and Project Management, and Project Management with all tracks. The key objective is to ensure that findings, model requirements, technical outputs, risks and testing requirements move between tracks and result in an actual change, decision, integration or validated output.

Key Dependencies

Major dependencies include: Analytics findings informing model interpretation and feature relevance; the Data Science candidate model and refined features being available to ML Engineering; downstream consumers confirming required prediction inputs/outputs and operational use and all integrated outputs being available for Week 7 end-to-end testing.

Risks and Limitations

Current limitations include moderate predictive performance, a need to validate the operating threshold against intervention capacity and false-positive/false-negative trade-offs, possible future data drift/generalisation issues, smaller sample sizes in some analytical segments, and the risk of overbooking congestion if risk scores are applied without capacity controls. One documented mitigation is a maximum 12% overbooking buffer, capped at two additional slots per four-hour clinical block.

Project Management Status

The PM workstream is coordinating integration asynchronously, maintaining the integration/dependency register, issue log, decision log and risk register, updating the project timeline based on actual progress, and preparing Week 7 testing coordination. The focus is on keeping all tracks aligned while allowing completed work to move into integration/review and testing readiness.

Week 7 Readiness

Week 7 will focus on Testing → Refinement → End-to-End Validation. Testing coordination should cover Analytics KPI/dashboard testing, Data Science model testing and validation, ML Engineering end-to-end pipeline testing, GenAI assistant testing, and cross-track end-to-end validation.

Overall Week 6 Summary

HealthConnect has progressed from separate Week 5 outputs toward integrated and validated workstreams. The strongest confirmed technical milestone is the ML Engineering pipeline validation with 10/10 tests passed, while Data Science has produced an improved candidate model and Data Analytics is supplying validated behavioural and KPI insights. The remaining priority is to close documented integration evidence, confirm outstanding dependencies and ensure all relevant outputs are ready for Week 7 testing.